

Best Practices in Python

Python is a widely-used scripting language. It emphasizes the code readability and has a syntax that allows programmers to express the concept in fewer lines than other languages. Hence its popularity among developers is increasing day by day. While using Python for developments people should adhere to some basic rules.

Structure of the Repository: Most developers don't pay much attention to the structure of the repository. They just set up a code repository and some version control. Which is not right, as it is a crucial part of the architecture. Again, the layout is not the everything but it has its own importance. Whenever someone new lands on the project it becomes easier for them to go through the project.

In Python, according to Kenneth Reitz, the product owner of Python at Heroku, the following components should exist in the repository:

- README.rst
- LICENSE
- setup.py
- requirements.txt
- sample/_init_.py
- sample/core.py
- sample/helpers.py
- docs/conf.py
- docs/index.rst
- tests/test_basic.py
- tests/test_advanced.py



An organised repository increases understandability of the project

- License: It is next most important thing after the source code. This file should contain license data and also the copyright claims. There are websites like choosealicense.com that help you choose one.
- README: There should be a basic README that will contain project description, outline of the basic functionality and the purpose of project.
- The Actual Module: This is the heart of the project. If the project contains a single file, then it should be placed in the root directory.
- Setup.py: This is the setup script which describes module distribution of the project to the distutils so that things operate correctly on the modules.
- Requirements.txt: This is the optional file can be used to specify development modules and other dependencies required for running the project properly.